

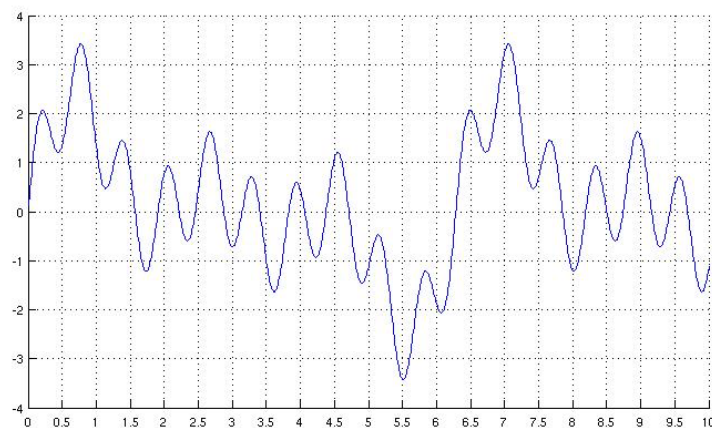
COMS20017 – Algorithms & Data

Problem Sheet MM03

1 – Using $\sin(2\pi nx)$, demonstrate the concept of superposition as follows (in Matlab or Python):

- first plot three sine functions over the range ± 3 in steps of 0.1 using $n=\{1/4, 1, 2\}$. Note, plots should appear in the same graph to give a better sense of what is happening.
- Now plot in a different colour the sum of all the sines above.
- Add more sine functions over the same range and repeat step (b).

2 – Based on your understanding of the Nyquist Sampling Rate theorem, what is a sufficient sampling rate for the signal below? Hint: the signal is composed of the summation of $\sin(x)$, $\sin(2x)$, $\sin(3x)$ and $\sin(10x)$.



3 – Determine which is an **even** and which is an **odd** function:

- $f(x) = 7x^3 - x$
- $f(x) = 3x^2 + 1$
- $f(x) = 3x^2 \sin(x)$
- $f(x) = \frac{3}{(-x)^4 - 4}$
- $f(x) = \cos(x) + 5x - 3$

4 – The period of the signal $x(t) = 10 \sin 12\pi t + 4 \cos 18\pi t$ is:

- $\pi/4$
- $1/6$
- $1/9$
- $1/3$
- $1/30$

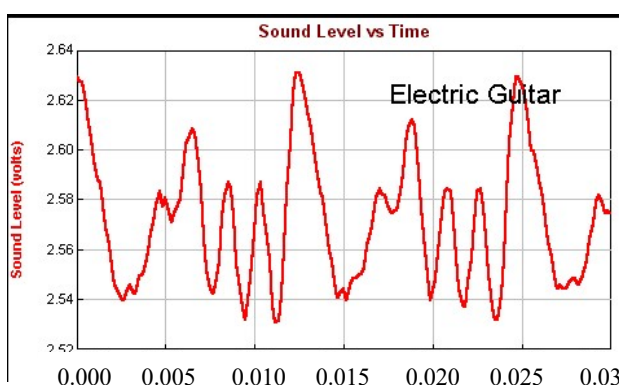
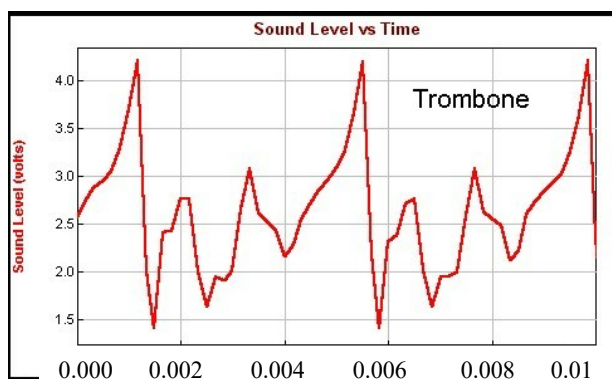
5 – The following gene sequence contains significant frequencies. Design two different symbolic encodings and in each case apply your encoding to extract some of these frequencies.

ACAGAGATACAGAGATACAG

6 – If the fastest oscillations that we want to measure are at 120 Hz, which of the following is the most reasonable sampling rate?

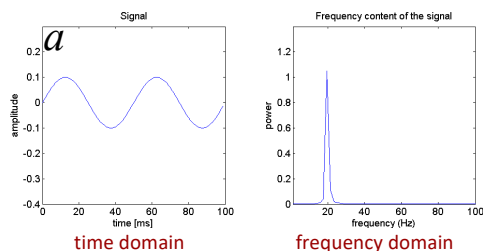
- a. 60 Hz
- b. 60 kHz
- c. anything over 0.00833 Hz
- d. 250 Hz
- e. 120 Hz

7 – The graphs below display the amplitude of the sound wave for a Trombone and an Electric Guitar as a function of time. The y-axis is the amplitude axis and the x-axis is the time axis. Notice that each one is plotted over a different length of time.

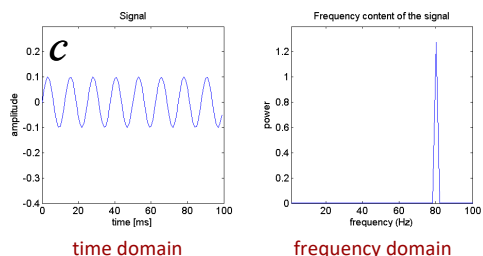
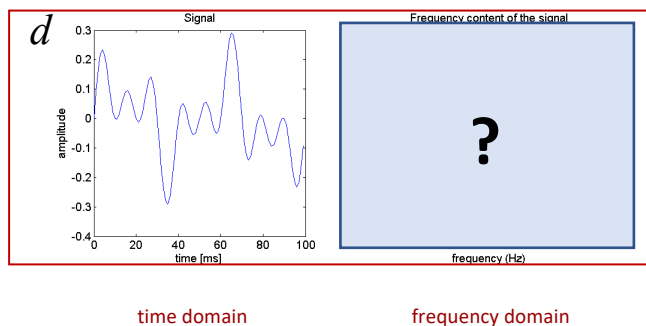
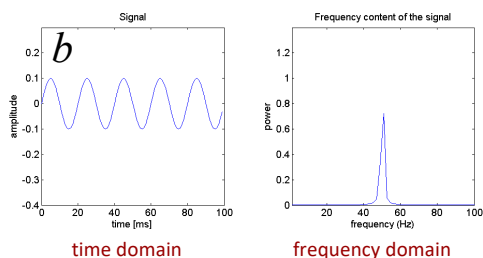


- (a) Mark the period of the signal for each instrument.
- (b) Approximately, how many periods are shown in these graphs for each instrument?
- (c) Approximately, what is the peak amplitude in each case?
- (d) Approximately, what is the frequency given the signal period in each case?
- (e) Which signal contains higher frequency information? Why?

8 – Consider the three signals a , b , and c below, and their addition d .



$$d = a + b + c$$



- What would the frequency of the signal d look like?
- How many oscillations per second does signal a have?
- How can you determine the frequency of signal c if you did not have the frequency domain plot of that signal?

9 – What are the two 1D filters that can replace the 2D filter for W , if they were applied consecutively? Do the same for X .

$$W = \frac{1}{9} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \quad X = \begin{pmatrix} 1 & 1 & -1 \\ 2 & 2 & -2 \\ 1 & 1 & -1 \end{pmatrix}$$

10 – In the Fourier Series of function $f(x) = x$, $-2 < x < 2$, the Fourier coefficient a_2 is equal to:

- 0
- 2
- 1
- 2